

KOTLAS, Russian Federation

WMO: 228870

Lat: 61.2395N

Lon: 46.6950E

Elev: 56

StdP: 100.66

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-29.4	-35.1	0.1	-32.7	-31.6	0.2	-29.4	8.4	-6.3	7.8	-5.8	1.3	330	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	28.6	20.6	26.5	19.5	24.7	18.4	21.6	27.0	20.5	25.2	19.3	23.3	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.6	24.5	18.6	13.6	23.1	17.5	12.6	21.8	63.3	27.2	59.1	25.3	55.1	23.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.8	6.1	-35.7	31.4	3.5	2.2	-38.2	32.9	-40.3	34.2	-42.2	35.4	-44.8	37.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-35.6	23.4	3.6	1.3	-38.2	24.4	-40.3	25.2	-42.3	25.9	-44.9	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-12.6	-11.5	-5.0	2.7	9.5	14.3	17.4	14.5	9.3	2.8	-4.7	-9.5
	DBStd	11.88	8.47	8.48	5.73	4.91	5.20	4.48	3.88	4.07	3.91	4.26	6.63	8.21
	HDD10.0	3415	701	602	466	224	75	11	7	59	227	439	604	
	HDD18.3	5903	959	835	724	469	276	136	66	133	272	482	689	862
	CDD10.0	619	0	0	0	5	61	140	229	145	37	3	0	0
	CDD18.3	68	0	0	0	0	3	15	36	13	1	0	0	0
	CDH23.3	707	0	0	0	2	54	171	368	109	2	0	0	0
	CDH26.7	166	0	0	0	0	9	40	87	29	0	0	0	0
	WSAvg	2.9	2.9	2.9	3.2	3.1	3.2	2.9	2.5	2.5	2.8	3.1	3.0	3.1
	PrecAvg	538	32	23	23	36	44	62	70	65	52	51	41	37
Precipitation	PrecMax	748	55	54	50	108	95	149	145	151	95	130	91	76
	PrecMin	402	8	2	2	3	15	15	19	17	13	5	12	10
	PrecStd	77	12	11	12	20	18	30	30	33	22	20	16	15
	PrecStd	77	12	11	12	20	18	30	30	33	22	20	16	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	1.6	2.7	8.4	21.8	28.0	30.9	31.5	30.0	23.0	16.1	8.3	4.1	
	0.4% MCWB	0.9	1.6	3.7	13.4	18.7	21.9	22.6	21.7	17.4	11.5	7.5	3.6	
	2% DB	0.8	1.5	5.4	16.3	24.7	27.4	29.0	26.1	19.6	12.4	5.1	2.1	
	2% MCWB	0.2	0.6	2.2	9.9	16.4	19.5	21.1	19.7	15.1	10.0	4.4	1.5	
	5% DB	-0.1	0.6	3.5	12.9	22.0	25.1	27.1	23.7	17.4	10.3	3.7	1.2	
	5% MCWB	-0.6	-0.1	1.3	7.6	14.5	18.3	20.3	18.5	13.7	8.7	3.1	0.6	
	10% DB	-1.8	-0.7	2.2	10.1	19.0	22.9	25.2	21.4	15.4	8.7	2.2	0.4	
	10% MCWB	-2.4	-1.4	0.5	5.7	12.9	16.8	19.1	17.2	12.8	7.5	1.7	-0.1	
	0.4% WB	1.1	2.0	4.3	13.5	19.7	22.5	23.4	22.6	17.8	12.3	7.4	3.6	
	0.4% MCDB	1.3	2.5	6.8	21.7	25.4	28.6	29.2	28.3	22.1	14.4	8.1	4.0	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	0.4	0.8	2.9	10.2	17.2	20.7	22.0	20.8	15.9	10.2	4.5	1.7	
	2% MCDB	0.8	1.3	4.7	15.7	23.1	25.9	27.5	25.3	18.6	11.9	4.9	2.0	
	5% WB	-0.7	-0.1	1.7	8.0	15.3	19.2	20.9	19.2	14.4	8.8	3.1	0.7	
	5% MCDB	-0.2	0.5	3.1	12.2	20.4	23.5	25.8	22.5	16.6	10.1	3.5	1.0	
	10% WB	-2.4	-1.4	0.7	6.3	13.4	17.7	19.8	17.7	13.0	7.5	1.8	-0.1	
	10% MCDB	-1.9	-0.7	2.0	9.5	18.4	22.0	24.4	20.6	15.2	8.7	2.1	0.4	
Mean Daily Temperature Range	MDBR	6.5	7.4	8.7	9.3	11.0	10.6	10.7	9.4	7.9	4.8	4.4	5.4	
	5% DB	5.7	5.1	8.7	14.0	15.7	14.3	13.9	12.6	11.3	7.9	4.4	3.9	
	MCDBR	5.7	4.8	6.2	8.7	8.5	7.2	6.9	6.7	7.0	5.7	4.2	3.8	
	5% WB	5.5	4.8	6.6	12.9	13.9	12.6	12.7	11.3	9.8	7.3	4.2	3.7	
	MCWBR	5.6	4.7	5.2	8.3	8.0	6.9	6.8	6.5	6.4	5.7	4.1	3.8	
	MCWBR	5.6	4.7	5.2	8.3	8.0	6.9	6.8	6.5	6.4	5.7	4.1	3.8	
Clear-Sky Solar Irradiance	taub	0.255	0.300	0.334	0.366	0.358	0.357	0.391	0.375	0.343	0.322	0.275	0.227	
	taud	2.011	2.058	2.064	2.195	2.432	2.462	2.370	2.395	2.468	2.362	2.109	1.950	
	Ebn at Noon	436	646	759	819	866	872	825	809	774	645	413	291	
	Edh at Noon	42	77	109	117	100	99	106	96	75	58	38	26	
All-Sky Solar Radiation	RadAvg	0.29	0.96	2.23	3.71	4.94	5.24	5.32	3.69	2.07	0.81	0.25	0.14	
	RadStd	0.04	0.22	0.35	0.44	0.35	0.54	0.47	0.40	0.25	0.10	0.05	0.02	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	+0.59	N/A	N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A	

Nomenclature: See separate page